Multi-Level On-Policy Preference Distillation for Small Language Models

Anonymous submission

Abstract

Alignment techniques enable large language models (LLMs) to generate outputs that better reflect human preferences and values. However, preference alignment remains challenging for small language models (SLMs) due to their limited capacity. Preference distillation offers a promising solution by transferring alignment knowledge from a preference-aligned teacher, yet existing methods often rely on offline data or incur substantial costs in on-policy learning through repeated response sampling or vocabulary-wide computation. We propose Multi-Level On-Policy Preference Distillation (MOPD), a unified framework that transfers preference knowledge from a teacher to an SLM using student-generated responses. MOPD performs distillation at both the sequence and token levels. At the sequence level, it derives a teacher-induced reward threshold through Bayesian classification, enabling preference supervision without multiple candidate responses. At the token level, it employs skewed reverse Kullback–Leibler divergence to provide fine-grained behavioral guidance on the student's current policy. Extensive experiments across three teacher–student configurations and multiple benchmarks show that MOPD consistently outperforms strong preference optimization and distillation baselines, achieving the best overall performance across all configurations. These results establish MOPD as an effective and general framework for on-policy preference distillation, paving the way for more efficient and accessible alignment of resource-constrained language models.

Introduction

Aligning large language models (LLMs) with human values and preferences has become a central objective in modern AI research. The goal of alignment is to guide models toward generating responses that are helpful, safe, and consistent with diverse and often complex human preferences and norms. Early approaches predominantly relied on Reinforcement Learning from Human Feedback (RLHF) (Christiano et al. 2017), while recent preference optimization methods, such as Direct Preference Optimization (DPO) (Rafailov et al. 2023) and its variants, provide more stable and computationally efficient alternatives by directly optimizing models from preference data. Despite their effectiveness, preference alignment has been studied primarily in large language models, which require substantial computational resources and are often available only as proprietary systems. These requirements limit their accessibility and deployment in resource-constrained settings. Small language models (SLMs), in contrast, offer a more practical alternative due to their lower computational and memory requirements, making them attractive for efficient and accessible deployment. However, effective preference alignment remains challenging for SLMs due to their limited model capacity. Moreover, alignment in SLMs can incur an **alignment tax**, where improvements in alignment are accompanied by degradation in general task performance (Bai et al. 2022).

Preference distillation (Gao et al. 2025) provides a promising paradigm for addressing these challenges by transferring alignment knowledge from a large, preference-aligned teacher to a smaller student model. Unlike conventional knowledge distillation for language models, which primarily focuses on transferring capabilities such as instruction following and general task performance, preference distillation aims to transfer the teacher's knowledge of human preferences, enabling the student to better distinguish between desirable and undesirable responses. Existing preference distillation methods can be broadly categorized into **off-policy** and **on-policy** approaches. Off-policy methods typically rely on preference data or teacher-generated responses that are constructed independently of the student's current policy. While computationally convenient, this training paradigm introduces a distribution mismatch between training and inference, as the student learns from a fixed external response distribution but generates responses from its own policy at inference time (Agarwal et al. 2024; Ko et al. 2024, 2025). On-policy preference distillation addresses this mismatch by deriving supervision directly from student-generated responses. Recent approaches such as Advantage-Guided Distillation for Preference Alignment (ADPA) (Gao et al. 2025) and Preference-Aligned Distillation (PAD) (Gu et al. 2025) follow this paradigm. ADPA derives token-level advantage signals from a DPO-trained teacher and its reference model, but requires vocabulary-wide computation at every token position, leading to computational costs that grow with both response length and vocabulary size. PAD instead samples multiple responses from the student and uses the teacher to

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evaluate their relative preferences, thereby reducing the distribution mismatch but incurring substantial generation and teacher-evaluation overhead. These limitations highlight the need for an efficient on-policy preference distillation framework that can leverage student-generated responses while avoiding both expensive vocabulary-level computation and repeated response sampling.

To address these challenges, we propose **Multi-Level On-Policy Preference Distillation (MOPD)**, a unified framework that transfers preference knowledge from a preference-aligned teacher to a smaller student at both the sequence and token levels. Our key insight is to formulate sequence-level preference distillation as a Bayesian classification problem. Rather than requiring multiple student responses to explicitly estimate a preference distribution, MOPD classifies each student-generated response by comparing its reward against a decision threshold induced by the teacher's reward distributions. Specifically, we model the reward distributions of preferred and dispreferred responses and derive a decision threshold by minimizing the risk of the corresponding Bayes classifier. This converts the teacher's preference knowledge into an efficient sequence-level supervision signal without requiring multiple candidate responses for each prompt. MOPD further performs token-level distillation on the same student-generated responses using skewed reverse KL (SRKL) (Ko et al. 2024), which emphasizes the probability mass associated with the student's current behavior while avoiding excessively aggressive policy updates. By combining sequence-level preference supervision with token-level distributional guidance, MOPD provides a unified mechanism for transferring both coarse-grained preference judgments and fine-grained behavioral knowledge from the teacher.

Our contributions are summarized as follows:

- **Multi-level on-policy preference distillation.** We introduce **MOPD**, a unified framework that distills preference knowledge from a preference-aligned teacher at both the sequence and token levels using student-generated responses, thereby mitigating the distribution mismatch inherent in offline preference distillation.
- **Teacher-induced preference threshold.** We formulate sequence-level preference distillation as a Bayesian classification problem and derive a principled decision threshold by minimizing the risk of the Bayes classifier over the teacher's preferred and dispreferred reward distributions. This enables efficient sequence-level preference supervision without requiring multiple student responses.
- **Empirical evaluation.** Extensive experiments across diverse teacher–student model scales and evaluation benchmarks demonstrate that MOPD consistently improves both preference alignment and general capabilities of small language models over strong preference optimization and preference distillation baselines.

Related work

Preference Alignment. Preference alignment steers language models toward outputs that better reflect human preferences and values. RLHF (Christiano et al. 2017) typically trains a reward model and then optimizes the policy with reinforcement learning algorithms such as PPO (Schulman et al. 2017), but incurs substantial training complexity. DPO (Rafailov et al. 2023) simplifies this process by directly optimizing the policy from preference pairs without an explicit reward model or reinforcement learning loop. SimPO (Meng, Xia, and Chen 2024) further adopts a reference-free, length-normalized reward based on the average response log probability, with an explicit margin to encourage stronger preference separation.

Preference Distillation. Preference distillation transfers alignment knowledge from a stronger teacher to a smaller student, offering an alternative to directly optimizing capacity-limited models. Existing methods broadly follow two paradigms: **off-policy** distillation, which relies on responses independent of the current student policy, and **on-policy** distillation, which uses student-generated responses. Off-policy methods learn from fixed preference data: DCKD (Gao et al. 2025) matches student and teacher output distributions on preferred and dispreferred responses, while TVKD (Kwon et al. 2025) uses the teacher's value function as a soft reward for fine-grained supervision. On-policy methods instead derive preference signals from student-generated responses: ADPA (Gao et al. 2025) provides token-level advantage supervision from a DPO-trained teacher and its reference model, but requires vocabulary-wide computation at every token position, with costs scaling with response length and vocabulary size. PAD (Gu et al. 2025) samples multiple student responses and uses the teacher to evaluate their preferences, incurring repeated generation and teacher-evaluation costs that grow with the number of candidates. Thus, existing on-policy methods remain computationally demanding, motivating a more efficient approach that leverages student-generated responses without vocabulary-wide computation or repeated sampling.

Methodology

Problem Setting and Notations

We formulate preference distillation as binary classification over pairwise response comparisons. Let $f_\theta : (x, y, y') \to [0, 1]$ denote a preference classifier, where x is a prompt, y, y' are candidate responses, and $s \in \{0, 1\}$ indicates whether $y \succ y'$ ($s = 1$) or $y' \succ y$ ($s = 0$). The Bayes-optimal classifier minimizes the expected classification risk:

$$f^* = \arg\min_\theta R(f_\theta), \qquad R(f_\theta) = \mathbb{E}\left[\ell(f_\theta, s)\right], \tag{1}$$

where we use the binary cross-entropy loss

$$\ell(f_\theta, s) = -s \log f_\theta(x, y, y') - (1-s) \log\left(1 - f_\theta(x, y, y')\right). \tag{2}$$

Following the Bradley–Terry model (Bradley and Terry 1952), we model the preference classifier by the reward difference:

$$\begin{aligned} f_\theta(x, y, y') = \sigma(r_\theta(x, y) - r_\theta(x, y'))\, P(s = 1) \\ + \sigma(r_\theta(x, y') - r_\theta(x, y))\, P(s = 0), \end{aligned} \tag{3}$$

where $r_\theta(x, y)$ is the reward function and $\sigma(\cdot)$ is the sigmoid function. The two terms correspond to the two preference

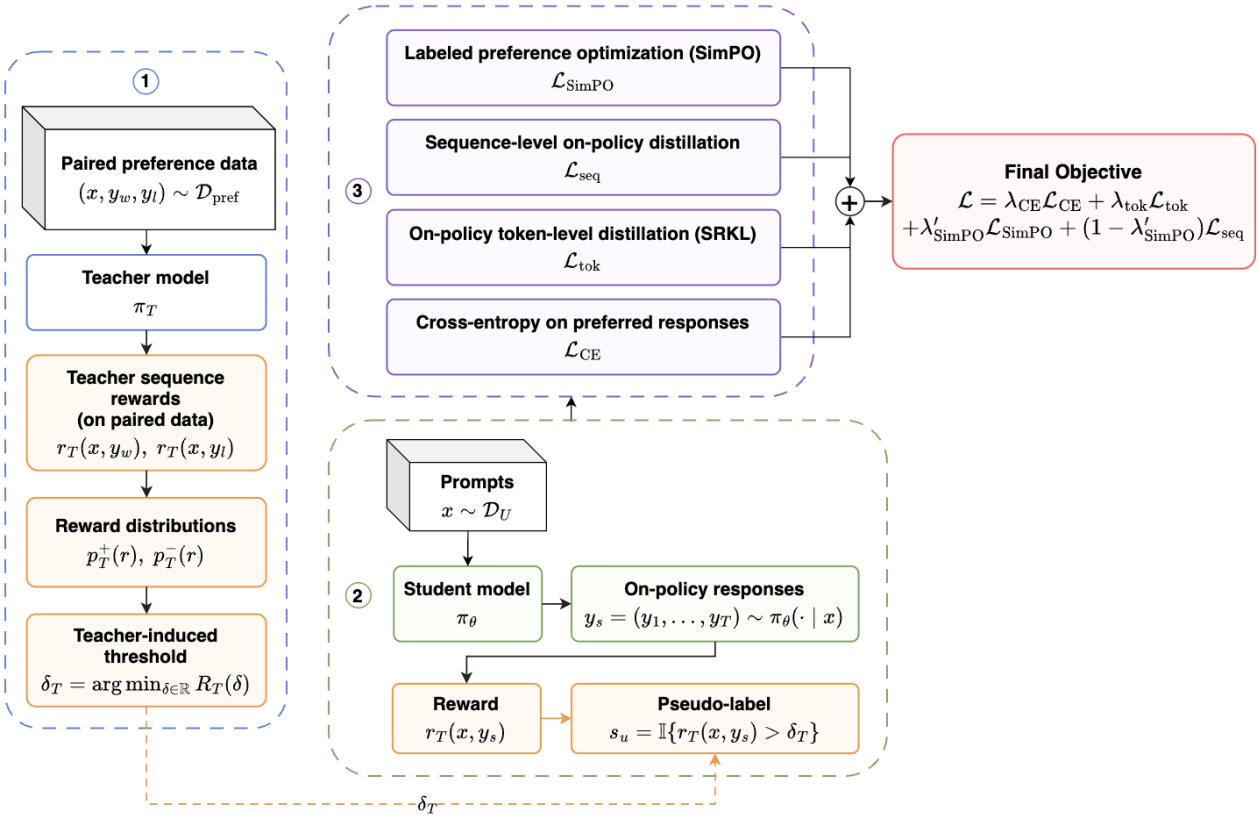


Figure 1: Overview of the proposed MOPD framework. The teacher derives a preference threshold from paired preference data, which is used to provide sequence-level preference supervision on student-generated responses. The same on-policy responses are further used for token-level SRKL distillation, and the resulting objectives are jointly optimized with cross-entropy supervision.

directions and are weighted by their respective class priors. To instantiate the reward function in the above preference model for language policies, we adopt the length-normalized sequence-level reward formulation of SimPO (Meng, Xia, and Chen 2024). Specifically, for a policy π, we define

$$r(x, y) = \frac{\beta}{|y|} \log \pi(y \mid x), \tag{4}$$

where β is a scaling factor and $|y|$ denotes the response length. Accordingly, the student and teacher rewards are given by

$$r_\theta(x, y) = \frac{\beta}{|y|} \log \pi_\theta(y \mid x), \tag{5}$$

$$r_T(x, y) = \frac{\beta}{|y|} \log \pi_T(y \mid x), \tag{6}$$

where π_θ and π_T denote the student and preference-aligned teacher policies, respectively. The length normalization reduces the dependence of the reward on response length, while the sequence log-probability provides a scalar measure of each policy's preference for the response.

Teacher-Guided threshold for Sequence-Level Distillation

We derive a teacher-induced decision threshold to transfer preference knowledge from the preference-aligned teacher to the student. Since the teacher is preference-aligned, its reward distributions over preferred and dispreferred responses capture the preference structure learned from the preference data. Given the preference dataset $\mathcal{D}_{\text{pref}}$, we denote the teacher reward distributions of preferred and dispreferred responses by $p_T^+(r)$ and $p_T^-(r)$, respectively. We then derive a decision boundary directly from these teacher-induced distributions. Specifically, for a threshold δ and preference label $s \in \{0, 1\}$, where $s = 1$ denotes the reward exceeds threshold, the corresponding Bayes risk is:

$$R_T(\delta) = P(s = 1) \int_{-\infty}^{\delta} p_T^+(r)\, dr + P(s = 0) \int_{\delta}^{\infty} p_T^-(r)\, dr. \tag{7}$$

The first and second terms correspond to misclassifying preferred and dispreferred responses, respectively. We define the optimal teacher-induced threshold as

$$\delta_T^* = \arg\min_{\delta \in \mathbb{R}} R_T(\delta). \tag{8}$$

This threshold provides a teacher-calibrated decision boundary for determining whether a student-generated response exhibits sufficiently preferred behavior.

Drawing inspiration from Semi-Supervised Preference Optimization (SSPO) (Lee et al. 2026) and applying it to our teacher reward distribution, we next establish the existence

of a teacher-induced reward threshold that separates winning and losing responses with high probability. Under mild distributional assumptions, sufficiently separated reward distributions admit a threshold that simultaneously lies above all losing rewards and below all winning rewards. The proof is provided in the Technical Supplement.

Theorem 1 (Existence of an Optimal Reward Threshold) *Consider i.i.d. teacher reward samples from losing responses, $\{r_T(x^{(i)}, y_l^{(i)})\}_{i=1}^{n_L}$, and winning responses, $\{r_T(x^{(j)}, y_w^{(j)})\}_{j=1}^{n_L}$. Assume that the two reward distributions are sub-Gaussian with means μ_l, μ_w and variance proxies σ_l^2, σ_w^2, respectively, with $\mu_w > \mu_l$. Then, for any $\alpha \in (0,1)$ and non-negative t_1, t_2 satisfying*

$$n_L \left(\exp\left(-\frac{t_1^2}{2\sigma_l^2} \right) + \exp\left(-\frac{t_2^2}{2\sigma_w^2} \right) \right) \leq \alpha, \tag{9}$$

there exists a threshold $\delta_T^* \in [\mu_l + t_1, \mu_w - t_2]$. such that

$$\mathbb{P}\left(\max_i r_\theta(x^{(i)}, y_l^{(i)}) \leq \delta_T^* \leq \min_j r_\theta(x^{(j)}, y_w^{(j)}) \right) \geq 1 - \alpha, \tag{10}$$

for all $i, j \in I(\mathcal{D}_{pref})$, where $I(\mathcal{D}_{pref})$ denotes the index set of instances in the preference dataset $\mathcal{D}_{pref}$. Thus, the result guarantees that, with probability at least $1-\alpha$, a threshold exists that simultaneously separates all losing and winning responses.

Theorem 1 provides a theoretical basis for using a reward threshold as a reliable decision boundary when the teacher assigns sufficiently separated rewards to winning and losing responses. Since the underlying teacher reward densities are unknown, we estimate them non-parametrically from the paired preference dataset $\mathcal{D}_{\text{pref}}$ using kernel density estimation:

$$\hat{p}_T^+(r) = \frac{1}{|\mathcal{D}_{\text{pref}}| h} \sum_{(x, y_w, y_l) \in \mathcal{D}_{\text{pref}}} K\left(\frac{r - r_T(x, y_w)}{h} \right), \tag{11}$$

$$\hat{p}_T^-(r) = \frac{1}{|\mathcal{D}_{\text{pref}}| h} \sum_{(x, y_w, y_l) \in \mathcal{D}_{\text{pref}}} K\left(\frac{r - r_T(x, y_l)}{h} \right), \tag{12}$$

where $K(\cdot)$ is a Gaussian kernel and h is the bandwidth. We then obtain the practical teacher-induced threshold by minimizing the empirical Bayes risk:

$$\hat{\delta}_T = \arg\min_{\delta \in \mathbb{R}} \hat{R}_T(\delta), \tag{13}$$

where

$$\hat{R}_T(\delta) = P(s=1) \int_{-\infty}^{\delta} \hat{p}_T^+(r)\, dr + P(s=0) \int_{\delta}^{\infty} \hat{p}_T^-(r)\, dr. \tag{14}$$

The resulting $\hat{\delta}_T$ defines a fixed, teacher-calibrated preference boundary. Given a prompt x, the current student samples $y \sim \pi_\theta(\cdot \mid x)$ and assigns the pseudo-label

$$\tilde{s} = \mathbb{I}\left\{ r_\theta(x, y) > \hat{\delta}_T \right\}. \tag{15}$$

We then reinterpret equation 3 with teacher-induced threshold to formulate sequence-level preference distillation objective as following:

$$\begin{aligned} \mathcal{L}_{seq} &= \frac{1}{N} \sum_{k=1}^{N} \ell\left(f_\theta(x^{(k)}, y^{(k)}), \tilde{s}_k \right) P_{\mathcal{D}}\left(s = \tilde{s}_k \right), \\ \tilde{s}_k &= \mathbb{I}\left\{ r_\theta(x^{(k)}, y^{(k)}) > \hat{\delta}_T \right\}. \end{aligned} \tag{16}$$

Here, $\mathcal{D}$ denotes the prompt dataset, and $P_{\mathcal{D}}(s=1)$ represents the prior probability that a student-generated response is preferred. This prior reflects the expected proportion of preferred responses and being fixed throughout training.

On-Policy Token-Level Preference Distillation

To provide token-level preference supervision on the student's current behavior, we perform distillation directly on student-generated trajectories. Given a prompt x, the student samples $y = (y_1, \ldots, y_S) \sim \pi_\theta(\cdot \mid x)$, and we define the on-policy distillation objective as

$$\mathcal{L}_{\text{tok}} = \mathbb{E}_{x \sim \mathcal{D},\, y \sim \pi_\theta(\cdot|x)} \left[\sum_{t=1}^{T} \mathcal{D}_{\text{SRKL}}^{(\alpha)}\left(\pi_T(\cdot \mid s_t) \,\|\, \pi_\theta(\cdot \mid s_t) \right) \right]. \tag{17}$$

where $s_t = (x, y_{<t})$ denotes the state visited at the t-th generation step and π_T is the preference-aligned teacher. We use skewed reverse KL (SRKL) (Ko et al. 2024) to compare the student and teacher distributions. Specifically, the SRKL divergence is defined as

$$\begin{aligned} &\mathcal{D}_{\text{SRKL}}^{(\alpha)}\left(\pi_T(\cdot \mid s_t) \,\|\, \pi_\theta(\cdot \mid s_t) \right) = \\ &D_{\text{KL}}\left(\pi_\theta(\cdot \mid s_t) \,\|\, (1-\alpha)\pi_T(\cdot \mid s_t) + \alpha \pi_\theta(\cdot \mid s_t) \right). \end{aligned} \tag{18}$$

where $\alpha \in [0, 1)$ controls the degree of interpolation between the teacher and student distributions.

Motivation for Skew Reverse KL. We adopt reverse KL (RKL) as the divergence for on-policy preference distillation. Unlike forward KL, RKL takes the expectation under the student distribution and therefore directly evaluates behaviors explored by the student. At the token level,

$$\mathcal{D}_{\text{RKL}}\left(\pi_T(\cdot \mid s_t) \,\|\, \pi_\theta(\cdot \mid s_t) \right) = \mathbb{E}_{y_t \sim \pi_\theta(\cdot|s_t)} \left[\log \frac{\pi_\theta(y_t \mid s_t)}{\pi_T(y_t \mid s_t)} \right]. \tag{19}$$

This student-centered formulation naturally provides a **pushing-down effect**: tokens that are frequently assigned probability by the student but receive substantially lower probability from the preference teacher model incur a large penalty, encouraging the student to reduce their probability. Such behavior is desirable for preference distillation, as it allows the student to suppress behaviors that are inconsistent with the preference teacher model. However, when the teacher preference model assigns a near-zero probability to a student-generated token, i.e., $\pi_T(y_t \mid s_t) \ll \pi_\theta(y_t \mid s_t)$, the resulting probability ratio in RKL can become arbitrarily large, leading to an unstable optimization signal. Thus, the

issue is not the pushing-down behavior itself, but rather the potentially unbounded magnitude of the resulting gradient.

To address this issue, we employ Skew Reverse KL (SRKL), which interpolates the teacher preference model distribution with the student distribution. The resulting ratio is

$$\frac{\pi_\theta(y_t \mid s_t)}{(1-\alpha)\pi_T(y_t \mid s_t) + \alpha\pi_\theta(y_t \mid s_t)} \tag{20}$$

Therefore, SRKL preserves the student-centered pushing-down behavior of RKL while preventing the corresponding ratio from becoming arbitrarily large when the preference-model probability is near zero. This interpolation yields a more stable optimization objective without discarding the desirable corrective behavior of reverse KL.

Overall MOPD Framework

MOPD jointly incorporates preference supervision from human-annotated data and on-policy supervision derived from student-generated responses. Let

$$D_{pref} = \left\{ \left(x^{(i)}, y_w^{(i)}, y_l^{(i)} \right) \right\}_{i=1}^{n_{pref}} \tag{21}$$

denote the preference dataset, where y_w and y_l denote the winning and losing responses, respectively, and let D denote the prompt dataset. Since each sample in D_{pref} explicitly identifies y_w and y_l as the preferred and dispreferred responses, respectively, the preference pair is fixed as $(x, y, y') = (x, y_w, y_l)$, yielding $s = 1$ for all samples in D_{pref}. Therefore, the expected classification risk on the preference dataset can be written as

$$\begin{aligned}
\mathbb{E}_{D_{\text{pref}}}\left[\ell(f_\theta, s)\right] &= \mathbb{E}_{D_{\text{pref}}}\left[\ell(f_\theta, 1) \mid s = 1\right] P_{D_{\text{pref}}}(s = 1) \\
&\quad + \mathbb{E}_{D_{\text{pref}}}\left[\ell(f_\theta, 0) \mid s = 0\right] P_{D_{\text{pref}}}(s = 0) \\
&= \mathbb{E}_{D_{\text{pref}}}\left[\ell(f_\theta, 1)\right] \\
&= \mathbb{E}_{D_{\text{pref}}}\left[-\log \sigma\left(r_\theta(x, y_w) - r_\theta(x, y_l)\right)\right].
\end{aligned} \tag{22}$$

Substituting the reward function in equation 5 into the above classification risk yields the SimPO objective

$$\begin{aligned}
\mathcal{L}_{\text{SimPO}} = -\mathbb{E}_{(x, y_w, y_l) \sim D_{pref}} \Bigg[\log \sigma \Bigg(& \frac{\beta}{|y_w|} \log \pi_\theta(y_w \mid x) \\
& - \frac{\beta}{|y_l|} \log \pi_\theta(y_l \mid x) - \Delta \Bigg) \Bigg].
\end{aligned} \tag{23}$$

here we add Δ, which is the target reward margin that encourages the winning response to receive a higher reward than the losing response.

To effectively combine labeled preference pairs and student-generated responses, MOPD employs an adaptive scheduler (Lee et al. 2026) rather than a fixed weighting scheme. Specifically, we dynamically adjust the coefficient λ' to implement a curriculum in which training initially prioritizes the more reliable human-annotated preference supervision, while progressively increasing the contribution of on-policy preference distillation. The overall training objective is

$$\mathcal{L}_{\text{MOPD}} = \lambda_{\text{CE}}\mathcal{L}_{\text{CE}} + \lambda_{\text{tok}}\mathcal{L}_{\text{tok}} + \lambda'\mathcal{L}_{\text{SimPO}} + (1-\lambda')\mathcal{L}_{\text{seq}}, \tag{24}$$

where

$$\lambda' = \max\left\{\lambda_{\min}, \lambda_0 \exp(-\rho\tau)\right\}. \tag{25}$$

Here, $\lambda' \in (0, 1]$ is the adaptive coefficient at training step τ ($1 \le \tau \le T$), $\rho > 0$ is the decay rate, and λ_0 is the initial value. We set $\lambda_0 = 1$ and

$$\lambda_{\min} = \frac{n_{pref}}{n_{pref} + n}, \tag{26}$$

where n_{pref} and n denote the numbers of preference pairs and prompts, respectively.

Experiments

Experimental settings

Baselines. We compare MOPD with two categories of baselines: *preference optimization* and *preference distillation*. The former includes **DPO** (Rafailov et al. 2023), which directly optimizes preferences without explicit reward modeling; **WPO** (Zhou et al. 2024), which reweights preference pairs according to their likelihood under the current policy to reduce distribution mismatch; **SimPO** (Meng, Xia, and Chen 2024), which uses a reference-free, length-normalized log-likelihood reward with a target margin; and **SSPO** (Lee et al. 2026), which combines limited labeled preference data with unpaired data. The latter includes **DCKD** (Gao et al. 2025), which distills teacher behavior on both preferred and dispreferred responses via dual KL constraints; **ADPA** (Gao et al. 2025), which provides fine-grained preference guidance using advantage signals derived from a preference-aligned teacher; **TVKD** (Kwon et al. 2025), which incorporates the teacher's value function as a soft reward through potential-based reward shaping; and **PAD** (Gu et al. 2025), which performs on-policy distillation by matching the teacher's preference distribution over student-generated responses.

Backbones and Training Data. We evaluate MOPD at two scales. At the small scale, we consider the teacher–student pair Qwen3-1.7B → Qwen3-0.6B (Yang et al. 2025). At the large scale, we consider Llama-3.1-8B → Llama-3.2-1B (Grattafiori et al. 2024) and Qwen3-8B → Qwen3-1.7B (Yang et al. 2025). All methods are initialized from the same SFT checkpoint, obtained by fine-tuning the respective student models on Deita10k-V0 (Liu et al. 2024), which contains approximately 10k high-quality instruction–response pairs. Preference distillation is subsequently performed on UltraFeedback Binarized (Cui et al. 2024), comprising over 63k binarized preference pairs. Following the original experimental setting of SSPO, we use only 10% of UltraFeedback Binarized and 10% of UltraChat-200k (Ding et al. 2023) as the unlabeled data. Accordingly, for MOPD, we use the same 10% subset of UltraFeedback Binarized and 10% of the prompts from UltraChat-200k to ensure a fair comparison.

Model	Method	HellaSwag	ARC	TruthfulQA	Winogrande	GSM8K	Average
Llama-3.1-8B → Llama-3.2-1B	DPO teacher	$83.05_{\pm 0.33}$	$52.30_{\pm 0.13}$	$57.37_{\pm 0.10}$	$78.22_{\pm 0.18}$	$52.77_{\pm 0.18}$	64.74
	SFT teacher	$81.85_{\pm 0.34}$	$51.20_{\pm 0.18}$	$53.59_{\pm 0.13}$	$78.45_{\pm 0.16}$	$47.84_{\pm 0.14}$	62.59
	SFT	$65.84_{\pm 0.42}$	$39.24_{\pm 0.14}$	$41.13_{\pm 0.34}$	$62.90_{\pm 0.11}$	$7.28_{\pm 0.71}$	43.28
	DPO	$66.94_{\pm 0.47}$	$39.25_{\pm 0.18}$	$44.57_{\pm 0.11}$	$62.12_{\pm 0.10}$	$\mathbf{7.88}_{\pm 0.79}$	44.15
	WPO	$67.44_{\pm 0.43}$	$40.70_{\pm 0.17}$	$42.45_{\pm 0.19}$	$62.27_{\pm 0.34}$	$6.22_{\pm 0.67}$	43.82
	SimPO	$67.40_{\pm 0.48}$	$\underline{40.83}_{\pm 0.10}$	$42.46_{\pm 0.14}$	$63.03_{\pm 0.15}$	$6.13_{\pm 0.77}$	43.97
	SSPO	$68.36_{\pm 0.48}$	$40.27_{\pm 0.14}$	$45.15_{\pm 0.16}$	$\underline{63.16}_{\pm 0.19}$	$6.55_{\pm 0.73}$	44.70
	PAD	$65.87_{\pm 0.41}$	$40.31_{\pm 0.13}$	$46.18_{\pm 0.29}$	$62.05_{\pm 0.29}$	$6.49_{\pm 0.61}$	44.18
	DCKD	$65.52_{\pm 0.43}$	$38.29_{\pm 0.18}$	$48.60_{\pm 0.16}$	$62.27_{\pm 0.53}$	$6.90_{\pm 0.13}$	44.32
	ADPA	$65.87_{\pm 0.41}$	$38.57_{\pm 0.22}$	$\mathbf{51.46}_{\pm 0.19}$	$62.27_{\pm 0.21}$	$6.88_{\pm 0.63}$	45.01
	TVKD	$\mathbf{67.64}_{\pm 0.49}$	$39.12_{\pm 0.14}$	$50.53_{\pm 0.16}$	$62.59_{\pm 0.17}$	$6.91_{\pm 0.62}$	$\underline{45.36}$
	MOPD	$\underline{67.59}_{\pm 0.47}$	$\mathbf{41.68}_{\pm 0.19}$	$\underline{51.26}_{\pm 0.11}$	$\mathbf{63.90}_{\pm 0.12}$	$\underline{7.86}_{\pm 0.57}$	**46.46**
Qwen3-8B → Qwen3-1.7B	DPO teacher	$80.60_{\pm 0.31}$	$66.55_{\pm 0.18}$	$61.52_{\pm 0.16}$	$75.93_{\pm 0.13}$	$80.89_{\pm 0.29}$	73.10
	SFT teacher	$79.30_{\pm 0.39}$	$66.04_{\pm 0.16}$	$58.96_{\pm 0.14}$	$75.69_{\pm 0.18}$	$79.53_{\pm 0.17}$	71.90
	SFT	$67.47_{\pm 0.33}$	$52.90_{\pm 0.12}$	$49.52_{\pm 0.11}$	$64.40_{\pm 0.14}$	$64.37_{\pm 0.15}$	59.73
	DPO	$68.44_{\pm 0.37}$	$53.84_{\pm 0.15}$	$52.54_{\pm 0.36}$	$64.01_{\pm 0.31}$	$65.25_{\pm 0.18}$	60.82
	WPO	$68.81_{\pm 0.33}$	$55.29_{\pm 0.17}$	$53.02_{\pm 0.19}$	$64.01_{\pm 0.17}$	$64.31_{\pm 0.16}$	61.71
	SimPO	$68.77_{\pm 0.39}$	$\underline{56.42}_{\pm 0.19}$	$54.03_{\pm 0.11}$	$64.77_{\pm 0.17}$	$62.22_{\pm 0.11}$	61.24
	SSPO	$\mathbf{69.36}_{\pm 0.31}$	$54.86_{\pm 0.18}$	$55.72_{\pm 0.12}$	$\underline{64.90}_{\pm 0.16}$	$64.64_{\pm 0.14}$	$\underline{61.90}$
	PAD	$67.56_{\pm 0.33}$	$54.82_{\pm 0.26}$	$54.94_{\pm 0.13}$	$64.17_{\pm 0.13}$	$64.29_{\pm 0.31}$	61.16
	DCKD	$66.75_{\pm 0.32}$	$54.92_{\pm 0.13}$	$54.26_{\pm 0.14}$	$64.64_{\pm 0.18}$	$\mathbf{66.49}_{\pm 0.14}$	61.41
	ADPA	$68.30_{\pm 0.39}$	$56.23_{\pm 0.19}$	$55.50_{\pm 0.18}$	$64.56_{\pm 0.14}$	$64.25_{\pm 0.16}$	61.77
	TVKD	$68.47_{\pm 0.30}$	$54.90_{\pm 0.11}$	$\underline{56.52}_{\pm 0.29}$	$64.40_{\pm 0.17}$	$64.37_{\pm 0.14}$	61.73
	MOPD	$\underline{69.15}_{\pm 0.32}$	$\mathbf{56.75}_{\pm 0.14}$	$\mathbf{57.23}_{\pm 0.11}$	$\mathbf{65.72}_{\pm 0.13}$	$\underline{66.05}_{\pm 0.14}$	**62.98**
Qwen3-1.7B → Qwen3-0.6B	DPO teacher	$68.44_{\pm 0.35}$	$53.84_{\pm 0.13}$	$52.54_{\pm 0.12}$	$64.01_{\pm 0.12}$	$65.25_{\pm 0.31}$	61.46
	SFT teacher	$67.47_{\pm 0.36}$	$52.90_{\pm 0.18}$	$49.52_{\pm 0.18}$	$64.40_{\pm 0.39}$	$64.37_{\pm 0.10}$	60.59
	SFT	$53.08_{\pm 0.33}$	$37.63_{\pm 0.16}$	$45.92_{\pm 0.39}$	$59.51_{\pm 0.16}$	$47.84_{\pm 0.18}$	49.09
	DPO	$53.74_{\pm 0.38}$	$39.08_{\pm 0.13}$	$46.92_{\pm 0.21}$	$59.43_{\pm 0.14}$	$52.46_{\pm 0.15}$	50.41
	WPO	$52.81_{\pm 0.37}$	$37.80_{\pm 0.14}$	$47.24_{\pm 0.18}$	$59.51_{\pm 0.18}$	$53.36_{\pm 0.19}$	50.17
	SimPO	$53.39_{\pm 0.33}$	$\underline{39.85}_{\pm 0.13}$	$47.69_{\pm 0.21}$	$59.14_{\pm 0.13}$	$50.04_{\pm 0.18}$	50.02
	SSPO	$54.26_{\pm 0.34}$	$37.70_{\pm 0.29}$	$46.82_{\pm 0.19}$	$59.69_{\pm 0.15}$	$52.27_{\pm 0.27}$	50.15
	PAD	$\mathbf{54.83}_{\pm 0.38}$	$38.12_{\pm 0.10}$	$46.15_{\pm 0.11}$	$\underline{60.21}_{\pm 0.23}$	$51.74_{\pm 0.37}$	50.21
	DCKD	$53.12_{\pm 0.29}$	$37.84_{\pm 0.21}$	$46.92_{\pm 0.17}$	$59.27_{\pm 0.11}$	$53.18_{\pm 0.17}$	50.01
	ADPA	$54.10_{\pm 0.32}$	$39.33_{\pm 0.12}$	$46.11_{\pm 0.18}$	$59.27_{\pm 0.17}$	$53.22_{\pm 0.12}$	50.50
	TVKD	$53.42_{\pm 0.39}$	$38.48_{\pm 0.19}$	$\mathbf{49.00}_{\pm 0.14}$	$59.98_{\pm 0.14}$	$\underline{54.44}_{\pm 0.16}$	$\underline{50.93}$
	MOPD	$\underline{54.51}_{\pm 0.39}$	$\mathbf{40.73}_{\pm 0.15}$	$\underline{48.06}_{\pm 0.18}$	$\mathbf{60.48}_{\pm 0.38}$	$\mathbf{54.49}_{\pm 0.11}$	**51.65**

Table 1: General capability results across three teacher–student configurations. The best and second-best student results within each configuration are shown in bold and underlined, respectively. Results are reported as mean with standard error.

Evaluation Metrics. We follow the evaluation protocol of the Hugging Face Open LLM Leaderboard (Beeching et al. 2023), implemented with the Language Model Evaluation Harness (Gao et al. 2021). We evaluate five benchmarks: ARC (Clark et al. 2018), HellaSwag (Zellers et al. 2019), Winogrande (Sakaguchi et al. 2020), TruthfulQA (Lin, Hilton, and Evans 2022), and GSM8K (Cobbe et al. 2021), covering science reasoning, commonsense reasoning, truthfulness, and mathematical reasoning.

Hyperparameters. We train all models for one epoch using the AdamW optimizer (Loshchilov and Hutter 2019) across 2 NVIDIA H200-141GB GPUs. For SFT, we use a global batch size of 16 and a learning rate of 2×10^{-4}. For MOPD, we use a global batch size of 64 and a learning rate of 5×10^{-6}. Unless otherwise specified, all other training settings are kept consistent across experiments. The random seed is globally set to 0

Main Results

Comparison with Preference Alignment and Preference Distillation Baselines. As shown in Table 1, MOPD consistently achieves the best overall performance among all evaluated student models across different teacher–student configurations. For the Llama-3.1-8B → Llama-3.2-1B setting, MOPD obtains an average score of 46.46, outperforming the strongest baseline, TVKD, by 1.10 points. The improvement is particularly pronounced on ARC, Winogrande, and GSM8K, where MOPD achieves gains of 2.56, 1.31, and 0.95 points, respectively. For the Qwen3-8B → Qwen3-1.7B setting, MOPD further achieves an average score of 62.98, surpassing SSPO by 1.08 points and consistently improving performance on ARC (+1.89), TruthfulQA (+1.51), Winogrande (+0.82), and GSM8K (+1.41). Notably, MOPD also remains competitive in the smaller Qwen3-1.7B → Qwen3-0.6B setting, achieving an average score of 51.65, which is the

Model	Method	HellaSwag	ARC	TruthfulQA	Winogrande	GSM8K	Average
Qwen3-8B → Qwen3-1.7B	**MOPD**	**69.15**	**56.75**	**57.23**	**65.72**	**66.05**	**62.98**
	w/o CE loss	68.04	53.67	52.50	65.19	65.81	61.042
	w/o SimPO loss	67.82	53.75	52.23	64.88	64.90	60.716
	w/o token level loss	67.80	53.07	51.88	65.27	64.67	60.538
	w/o sequence level loss	67.77	54.18	51.66	64.80	64.52	60.586

Table 2: Ablation study of MOPD components on the Qwen teacher–student configuration.

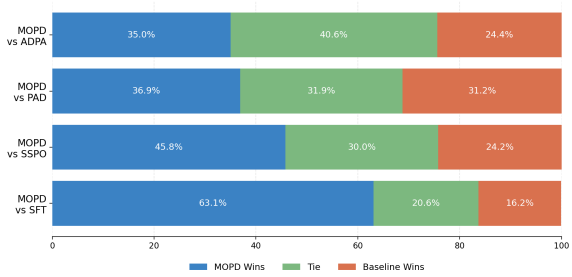


Figure 2: MOPD vs baselines on MT-Bench

best result among all student models. These results demonstrate that MOPD can effectively transfer preference knowledge from aligned teachers to substantially smaller models while maintaining strong general-purpose capabilities across diverse reasoning and knowledge-intensive benchmarks.

Comparison of Response Quality on MT-Bench. Beyond standard benchmark accuracy, we further evaluate the quality of generated responses on MT-Bench (Zheng et al. 2023) using pairwise comparisons, as shown in Figure 2. MOPD achieves competitive or superior win rates against all evaluated student baselines, providing a direct measure of its ability to improve response quality. In particular, MOPD wins against TVKD in 25.0% of comparisons and against PAD in 36.9%, while achieving a substantially higher win rate of 63.3% against the SFT baseline. These results complement the general-capability evaluation and indicate that the improvements brought by MOPD extend beyond benchmark scores to the quality and preference alignment of generated responses.

Ablation Study

Influence of Different Training Objectives. To investigate the contribution of individual training objectives to MOPD, we conduct an ablation study on the Qwen3-8B → Qwen3-1.7B teacher–student configuration. Specifically, we remove each loss component from the full MOPD objective while keeping all other components unchanged. The evaluated variants are as follows:

- **w/o CE Loss:** We remove the cross-entropy loss from the training objective, eliminating the direct supervision from the ground-truth response tokens.
- **w/o SimPO Loss:** We remove the labeled preference loss based on SimPO, thereby excluding direct preference supervision from the paired preference data.
- **w/o Token-Level Loss:** We remove the token-level distillation objective, preventing MOPD from exploiting fine-grained, token-level preference signals from student-generated responses.
- **w/o Sequence-Level Loss:** We remove the sequence-level distillation objective, eliminating the teacher-guided sequence-level preference supervision based on the estimated preference threshold.

As shown in Table 2, the full MOPD model achieves the highest average score of 62.98, while removing any individual component consistently degrades performance. Removing the CE, SimPO, token-level, and sequence-level losses reduces the average score to 61.04 (-1.94), 60.72 (-2.26), 60.54 (-2.44), and 60.59 (-2.39), respectively, with the largest drops resulting from removing the two multi-level distillation objectives. A similar trend is observed on TruthfulQA, where the full model achieves 57.23, compared with 52.50, 52.23, 51.88, and 51.66 for the corresponding ablated variants. These results demonstrate that the training objectives provide complementary supervision, with sequence-level distillation offering coarse-grained guidance through the teacher-induced preference boundary and token-level distillation providing fine-grained correction along student-generated trajectories, both of which are important for effectively transferring preference-related behavior.

Conclusion

We introduced **Multi-Level On-Policy Preference Distillation (MOPD)**, a framework that transfers preference knowledge from aligned teacher LLMs to small language models using student-generated trajectories. MOPD combines a teacher-induced reward threshold for sequence-level preference supervision with skewed reverse KL distillation for token-level behavioral correction, enabling effective on-policy preference transfer without extensive multi-response sampling. Extensive experiments across multiple teacher–student configurations show that MOPD consistently outperforms existing preference optimization and distillation baselines while preserving general capabilities. Ablation studies further confirm the complementary contributions of the sequence- and token-level objectives. These results demonstrate the potential of multi-level on-policy supervision for efficient preference alignment of resource-constrained language models.

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